Question 1 (25 marks) – CLO1: C3

Consider the ethical case in Scenario Q1. Use the Guidelines for Code of Professional Conduct to support your answers.

a) Identify the key points of the issue.

[6 marks]

b) Identify and explain the section(s) and/or subsection(s) of the Code of Professional Conduct relevant to the case under study.

[10 marks]

c) Is it ethical for Engineer X to seal plans that have not been prepared by him, or which he has not checked and reviewed in detail? Explain why.

[9 marks]

Scenario Q1

As the Chief Engineer of one of the leading integrated engineering services companies in Malaysia, Engineer X, a professional engineer, is responsible to approve and sign the engineering documents or plans that are prepared by the professional engineers under his direct supervision in the company. There are also occasions whereby Engineer X affixes his signature to plans or documents prepared by the graduate engineers reporting to him.

The company has gained confidence of clients by completing complex projects with innovative methods and enhanced equipment. With great business expansion of the company over the years, Engineer X needs to coordinate a large number of projects at a time. It is therefore challenging for him to thoroughly review and check each technical segment of all projects prior to his approval. Engineer X believes that the professional and graduate engineers in his team are competence and capable of fulfilling the requirements of the engineering design and specifications, hence detailed checking of the design is not necessary.

The clients of the company often consult Engineer X for the project information such as the design requirement, concept and updates on the progress of the work. Engineer X will attend to all these inquiries accordingly.

PAST EXAM PAPER

Question 2 (25 marks) – CLO1: C3

Consider the ethical case in Scenario Q2. Use the Guidelines for Code of Professional Conduct to support your answers.

a) Identify the key points of the issue.

[8 marks]

b) Identify and explain the section(s) and/or subsection(s) of the Code of Professional Conduct relevant to the case under study.

[10 marks]

c) What should Engineer Z do? Please provide reasonings.

[7 marks]

Scenario Q2

Engineer Z is a professional engineer specializes in digital signage solutions. In addition to his main responsibility of design and integration of digital signage solutions, he provides technical direction or supervision to junior engineers and other technical personnels at clients' site. The company in which Engineer Z is working for also manufacture outdoor and indoor Light Emitting Diode (LED) displays in their in-house factory.

One day, one of the company's clients reported that three large size outdoor LED displays had stopped working in his shopping mall approximately two months after commissioning. Technicians from the site highlighted that the serial numbers on the three LED displays were different from the remaining twelve LED displays installed for the shopping mall. Upon inspection, Engineer Z found that the three large LED displays were from the production batch which did not pass the quality control checking during manufacturing. Engineer Z would like to share this finding to the client but was aware that the client is eligible to seek for replacement units at no additional cost if he reveals the information. On the contrary, his company would be able to charge the client for replacement units if he puts the blame on improper handling and inadequate maintenance activities at site for the LED displays.

Engineer Z would like to be truthful in his inspection report to the client as he has been a loyal customer to the company for many years. However, Engineer Z was well-aware that his manager would want him to mitigate the issue with minimum losses to the company.

PAST EXAM PAPER

Question 3 (25 marks) – CLO3: C4

As an engineer, you have been tasked to collaborate with a utility company to evaluate the potential of an alternative energy as supplement or replacement for fossil fuels. The selected alternative energy will be evaluated against coal power plant. The selection of an alternative energy for evaluation, namely solar, wind, hydro and nuclear is based on the last digit of your student ID. For example, if your student ID is EEE1808252, the last digit is "2". The alternative energy will be determined from the last digit as follows: 0, 1 and 2 for solar energy; 3 and 4 for wind energy; 5, 6 and 7 for hydro energy; and 8 and 9 for nuclear energy.

a) Evaluate the feasibility of the selected alternative energy against coal power plant based on the selection criteria shown in Table Q2.

[16 marks]

b) Do you think the selected alternative energy can address the issues of security and reliability of supply, and over-dependence on fossil fuels in your country? Justify your answers.

[9 marks]

Table Q2 Criteria for alternative energy selection

Main Criteria	**Sub-criteria**
C1: Quality of the Energy Source	C11: Sustainability
C2: Socio-political	C21: Social acceptance
C3: Economic	C31: Implementation cost C32: Economic value
C4: Technological	C41: Continuity and predictability of the performance C42: Risk C43: Local technical knowledge
C5: Environmental	C51: Pollutant emission C52: Land requirement C53: Requirement for waste disposal

PAST EXAM PAPER

Question 4 (25 marks) – CLO3: C4

In the United Nations 2030 Agenda for Sustainable Development, climate change was addressed as "one of the greatest challenges of our time" and "its adverse impacts undermine the ability of all countries to achieve sustainable development" became a pressing concern.

a) Discuss the implications of climate change or global warming to the society and environment.

[9 marks]

b) Explain on the relationship between climate change and sustainable development.

[6 marks]

c) Explain the possible approaches in engineering practice to effectively meet the "Sustainable Development Goal 7: "Ensure access to affordable, reliable, sustainable and modern energy for all" and "Sustainable Development Goal 13: Take urgent action to combat climate change and its impacts".

[10 marks]

- END OF PAPER -

PAST EXAM